Logging & Monitoring Specification

Ref: 5.1
Product: iConnect Plus
Organisation: Cache Digitech
Classification: Internal — Confidential
Version: 2.0
Date: 2026-11-03
Deployment model: On-premises (Cache Digitech premises) · VPN for remote access
Audience: IT operations, SOC, platform engineering, security assessors

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1. Purpose

Specify **what** iConnect Plus logs, **where** logs are stored on-premises, **retention (≥180 days)**, and **monitoring** hooks for the internal Docker Compose deployment.

2. Log retention policy

Requirement	Target
Minimum retention	**≥ 180 days**
Storage location	**India — Cache Digitech premises** (NAS, log host, or aggregated store)
Public cloud log SaaS as primary	**Not used** in this model

Implementation evidence: Template / Pending sign-off (log path, rotation, archival job).

3. Log sources

Source	Container / component	Examples
API access & errors	`erp-api`	HTTP status, route, correlation ID
Auth events	`erp-api`	Login success/failure, lockout (no passwords)
Worker jobs	`erp-celery-worker`	Task failures, outbox drain
Web	`erp-web`	SSR errors, proxy failures
Edge	`erp-nginx`	Access log, TLS errors
Database	`erp-postgres`	**Template / Pending sign-off** (optional pg log)
Host OS	Docker host	SSH, sudo, kernel

4. Security logging rules

Rule	Detail
No secrets in logs	Passwords, tokens, Fernet plaintext, full JWT
PII minimization	Prefer user UUID over email where possible
Encrypted fields	Never log values from `ENCRYPTED_FIELD_INVENTORY`
Failed auth	Log count + source IP + user identifier (not password)
Privileged actions	Audit engine + application audit where configured

Reference: **2.1 Application Security Requirements** (LOG-* items).

5. Log format & correlation

Field	Requirement
Timestamp	UTC ISO-8601
Correlation / request ID	Propagate across API → worker where applicable
Tenant context	When authenticated
Severity	DEBUG disabled in production target

Structured JSON logging: **Template / Pending sign-off** (partial today — standardization target).

6. Aggregation & rotation

Step	Procedure
Collect	Docker logging driver → host path or agent
Rotate	logrotate daily/size — **Template / Pending sign-off**
Archive	Compress to NAS; delete after 180+ days
Integrity	**Template / Pending sign-off** (WORM / append-only)

7. Monitoring & alerting

Signal	Tool / route	Alert owner
API health	`/health` or platform SLO routes	Platform on-call
Container down	Docker / host monitor	IT
Disk full	Host metrics	IT
Failed login spike	Log query / SIEM	**Template / Pending sign-off**
Celery queue depth	RabbitMQ management	Platform
Backup failure	Backup job log	IT

SIEM product: Template / Pending sign-off

8. Access to logs

Role	Access
Platform on-call	Read API/worker/nginx logs
IT / SOC	Full host + container
Developers	Non-prod only (default)
Auditors	Read-only export under NDA

Remote access to log systems: **VPN required**.

9. Compliance mapping

Obligation	Control
CERT-In incident evidence	5.4 + retained logs
DPDP security safeguards	Access control on logs
Internal audit	180-day window

10. Related documents

Ref	Document
5.4	Incident Response Plan
5.6	Operations Runbook
4.3	Data Retention & Deletion Schedule
2.1	Application Security Requirements

11. Change log

Date	Change	By
2026-11-03	v2.0 ≥180d on-prem logging spec	Platform

IT sign-off: Template / Pending sign-off